MILITARY SPECIFICATION SHEET

PIN-RIVET, THREADED, TYPE III (100° FLUSH CROWN SHEAR HEAD), CLASS 5
(ALLOY STEEL, 108 KSI Fsu AND 180-200 KSI Ftu)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the pin-rivet, threaded described herein shall consist of this document and the issue in effect of MIL-P-87988.

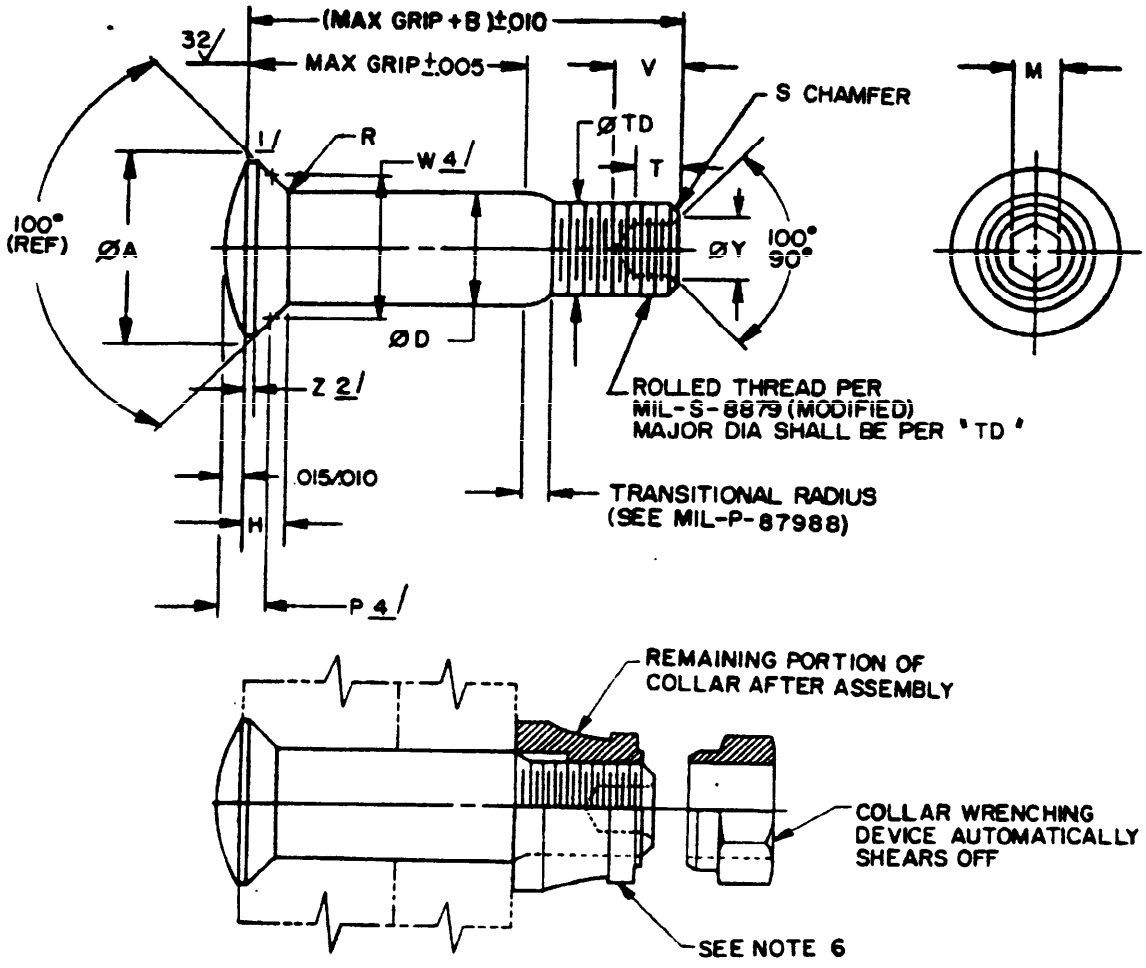


FIGURE 1. PIN-RIVET, THREADED, TYPE III (100° FLUSH CROWN SHEAR HEAD), CLASS 5 (ALLOY STEEL, 108 KSI Fsu AND 180-200 KSI Ftu)

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NOTES:

1. Dimensions in inches to be met after application of final coating or surface finish, unless otherwise specified.
1/ 2. Head edge out of roundness shall not exceed "F".
3. Concentricity: conical surface of head to "D" diameter within .005 FIM.
4. "H" dimensioned from maximum "D" diameter.
2/ 5. Curved or flat edge manufacturer's option.
3/ 6. Evidence of broken edge across points.
7. While this part is designed to be used in interference fit applications (because it has a transitional radius) it can also be used in noninterference or clearance applications. It is similar to HL619. "HL" is a registered trademark of Hi-Shear Corporation.
8. Hole preparation in accordance with NAS618. (Information only)
4/ 9. All head dimensions shall comply with requirements of TABLE I of this slash sheet and TABLE X of procurement specification.

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TABLE I. DASH NUMBERS AND DIMENSIONS

DIA NO.	NOM SIZE	THREAD (MODIFIED)	ØA	B REF	ØD	ØTD	1/ F	H	2/ R RAD	2/ Z	S CHAMFER REF	SOCKET M HEX	SOCKET T DEPTH	SOCKET V MAX DRILL DEPTH	SOCKET ØY	DOUBLE SHEAR POUNDS MIN	TENSION POUNDS MIN
5	5/32	.1640-32 UNJC-3A	.2612 .2564	.312	.1635 .1625	.1595 .1570	.004	.0408 .0388	.025 .015	.010 .005	.031 x 45°	.0801 .0791	.100 .080	.140	3/	4,560	2,000
6	3/16	.1900-32 UNJF-3A	.3016 .2966	.325	.1895 .1885	.1840 .1810	.005	.0470 .0449	.030 .020	.015 .005	.031 x 45°	.0806 .0791	.100 .080	.145	.119 .104	6,125	2,640
8	1/4	.2500-28 UNJF-3A	.3948 .3898	.395	.2495 .2485	.2440 .2410	.006	.0610 .0589	.030 .020	.015 .005	.031 x 45°	.0967 .0947	.110 .090	.160	.142 .122	10,600	4,600
10	5/16	.3125-24 UNJF-3A	.4739 .4689	.500	.3120 .3110	.3060 .3020	.007	.0679 .0658	.040 .030	.015 .005	.047 x 45°	.1295 .1270	.130 .110	.200	.180 .160	16,600	7,100
12	3/8	.3750-24 UNJF-3A	.5604 .5554	.545	.3745 .3735	.3680 .3640	.008	.0780 .0759	.040 .030	.015 .005	.047 x 45°	.1617 .1582	.160 .140	.239	.217 .197	23,900	9,950
14	7/16	.4375-20 UNJF-3A	.6680 .6620	.635	.4370 .4360	.4310 .4260	.009	.0969 .0944	.050 .040	.022 .005	.047 x 45°	.1930 .1895	.190 .170	.278	.253 .233	32,500	14,000
16	1/2	.5000-20 UNJF-3A	.7540 .7480	.685	.4995 .4985	.4930 .4880	.085	.1068 .1043	.050 .040	.022 .005	.047 x 45°	.2242 .2207	.220 .200	.323	.289 .269	42,400	17,800
18	9/16	.5625-18 UNJF-3A	.8380 .8310	.770	.5615 .5605	.5550 .5500	.125	.1114 .1115	.050 .040	.025 .005	.063 x 45°	.2555 .2520	.260 .240	.371	.326 .306	53,700	19,000

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REQUIREMENTS:

1. Material. Alloy steel (UNS G43406) in accordance with MIL-S-5000, (UNS G41400) MIL-S-5626 or (UNS G87400) MIL-S-6049.

2. Protective coating and surface treatment

   All pins shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2.

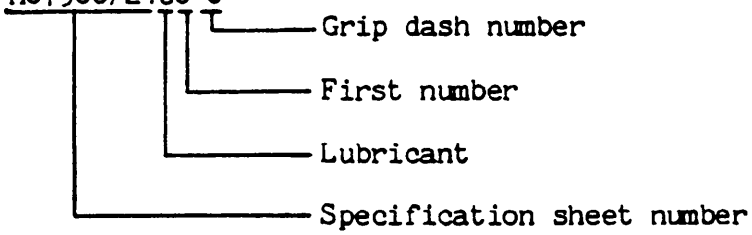
   Lubricant:
   C - Cetyl alcohol lubricant in accordance with MIL-L-87132 Type I, Grade B.

3. Heat treatment. 108,000 psi shear minimum (180,000-200,00 psi tensile) in accordance with MIL-H-6875.

4. Surface texture. Unless otherwise specified, all surfaces shall have a 125 microinch finish.

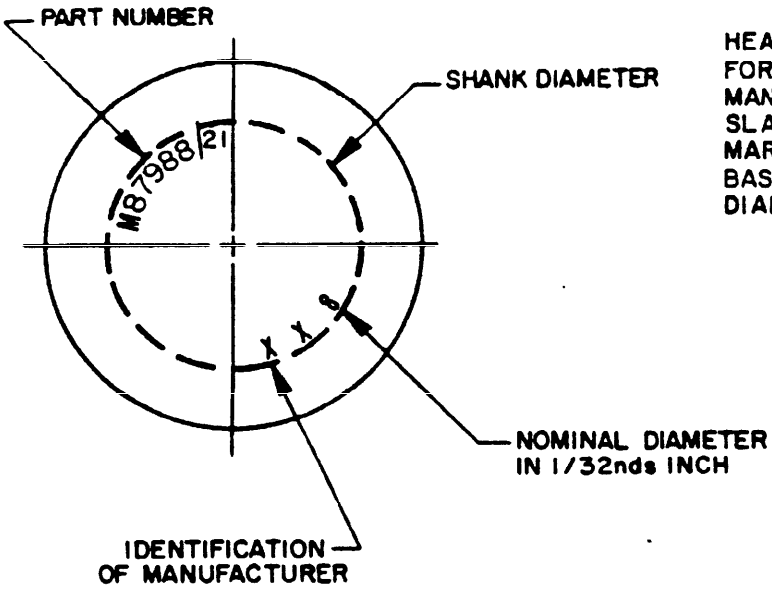
5. Part number. The part number consists of the letter "M", the basic number of this specification sheet, followed by a lubricant code letter, the first number indicats nominal diameters in thirty-seconds of an inch taken from TABLE I, the grip dash number indicates maximum grip length in sixteenths of an inch taken from TABLE II.

   Example: M87988/21C8-8



M87988/21C8-8 indicates: Pin-Rivet, Threaded, Type III, (100° Flush Crown Shear Head), Class 5 (Alloy Steel, 108 KSI Fsu and 180-220 KSI Ftu); cadmium plated; Cetyl Alcohol lubricant; 8/32 nominal diameter; 8/16 maximum grip length.

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HEAD MARKING SHALL BE .010 MAXIMUM DEEP. FOR DASH 5,6 AND 8 SIZES ONLY, MARK MANUFACTURER IDENTIFICATION AND M PLUS SLASH SHEET (M21). FOR DASH 10 AND ABOVE, MARK MANUFACTURER IDENTIFICATION PLUS BASIC FULL PART NUMBER (M87988/21-) AND DIAMETER.

TABLE II. GRIP AND LENGTH DIMENSIONS

GRIP DASH NO.	MAX GRIP ±.005	LENGTH ±.010 a/							
		DIAMETER DASH NUMBER AND THREAD SIZE							
		5 .1640-32	6 .1900-32	8 .2500-28	10 .3125-24	12 .3750-24	14 .4375-20	16 .5000-20	18 .5625-18
1	.062	.375							
2	.125	.437	.450	.520	.625	.670	.760	.810	
3	.188	.500	.513	.583	.688	.733	.823	.873	.958
4	.250	.562	.575	.645	.750	.795	.885	.935	1.020
5	.312	.625	.638	.708	.813	.858	.948	.998	1.082
6	.375	.687	.700	.770	.875	.920	1.010	1.060	1.145
7	.438	.750	.763	.833	.938	.984	1.073	1.123	1.218
8	.500	.812	.825	.895	1.000	1.045	1.135	1.185	1.270
9	.562	.875	.888	.958	1.063	1.108	1.198	1.248	1.332
10	.625	.937	.950	1.020	1.125	1.170	1.260	1.310	1.385
11	.688	1.000	1.013	1.083	1.188	1.233	1.323	1.373	1.450
12	.750	1.062	1.075	1.145	1.250	1.295	1.385	1.435	1.520
13	.812	1.125	1.138	1.208	1.313	1.358	1.448	1.498	1.582
14	.875	1.187	1.200	1.270	1.375	1.420	1.510	1.560	1.645
15	.938	1.250	1.263	1.333	1.438	1.483	1.573	1.623	1.708
16	1.000	1.312	1.325	1.395	1.500	1.545	1.635	1.685	1.770
17	1.062	1.375	1.388	1.458	1.563	1.608	1.698	1.748	1.832
18	1.125	1.437	1.450	1.520	1.625	1.670	1.760	1.810	1.895
19	1.188	1.500	1.513	1.583	1.688	1.733	1.823	1.873	1.958
20	1.250	1.562	1.575	1.645	1.750	1.795	1.885	1.935	2.020
21	1.312	1.625	1.638	1.708	1.813	1.858	1.948	1.998	2.082
22	1.375	1.687	1.700	1.770	1.875	1.920	2.010	2.060	2.145
23	1.438	1.750	1.763	1.833	1.938	1.983	2.073	2.123	2.208
24	1.500	1.812	1.825	1.895	2.000	2.045	2.135	2.185	2.270
25	1.562	1.875	1.888	1.958	2.063	2.108	2.198	2.248	2.332
26	1.625	1.937	1.950	2.020	2.125	2.170	2.260	2.310	2.395
27	1.688	2.000	2.013	2.083	2.188	2.233	2.323	2.373	2.458
28	1.750	2.062	2.075	2.145	2.250	2.295	2.385	2.435	2.520
29	1.812	2.125	2.138	2.208	2.313	2.358	2.448	2.498	2.582
30	1.875	2.187	2.200	2.270	2.375	2.420	2.510	2.560	2.645
31	1.938	2.250	2.263	2.333	2.438	2.483	2.573	2.623	2.708
32	2.000	2.312	2.325	2.395	2.500	2.545	2.635	2.685	2.770

a/ Nominal length equals maximum grip plus "B". See FIGURE 1.

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Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Army - GL
DLA - IS
NSA - NS

User activities:

Preparing activity:
Air Force - 82

Agent:
DLA - IS

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